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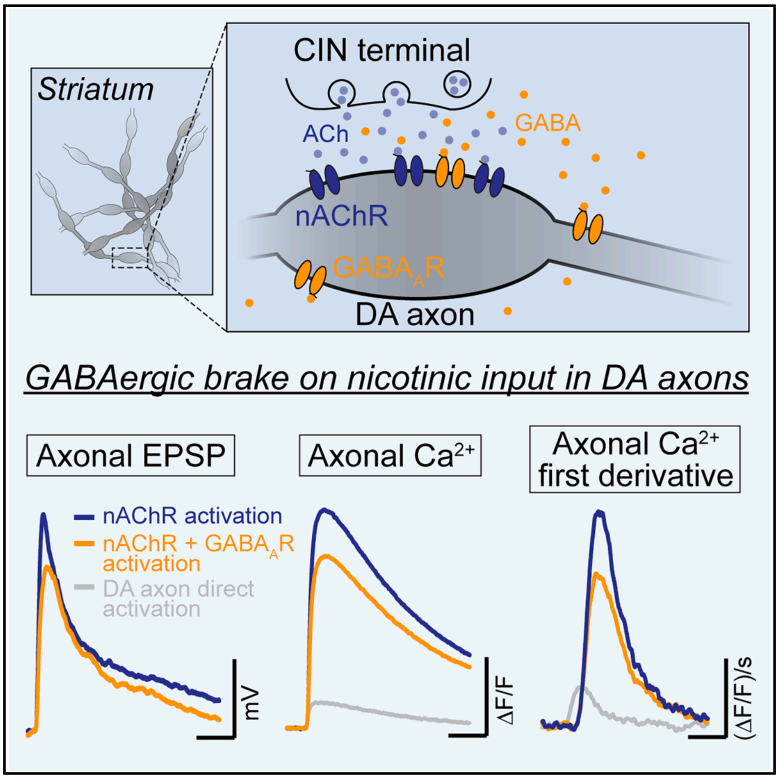
Presynaptic $GABA_A$ receptors control integration of nicotinic input onto dopaminergic axons in the striatum

Graphical abstract



Highlights

- Axonal $GABA_A$Rs provide a brake on nicotinic drive to striatal dopaminergic axons
- Derivative Ca^{2+} signals enable separate analysis of intrinsic and circuit-driven axon activity
- Picrotoxin, but not gabazine, blocks axonal nAChRs at concentrations used to inhibit $GABA_A$Rs
- Dopaminergic axons perform integration of inhibition and excitation to shape output

Authors

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In brief

Brill-Weil et al. show that axonal $GABA_A$ receptors regulate excitatory nicotinic input onto the axons of dopaminergic neurons. This regulation alters integration of the subthreshold nicotinic input to impact axonal action potential output in the striatum.

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Presynaptic $GABA_A$ receptors control integration of nicotinic input onto dopaminergic axons in the striatum

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SUMMARY

Axons of dopaminergic neurons express gamma-aminobutyric acid type-A receptors ($GABA_A$Rs) and nicotinic acetylcholine receptors (nAChRs), which are positioned to shape striatal dopamine release. We examine how interactions between $GABA_A$Rs and nAChRs influence dopaminergic axon excitability. Axonal patch-clamp recordings reveal that potentiation of $GABA_A$Rs by benzodiazepines suppress dopaminergic axon responses to cholinergic interneuron transmission. In imaging experiments, we use the first temporal derivative of axonal calcium signals to distinguish between direct stimulation of dopaminergic axons and nAChR-evoked activity. Inhibition of $GABA_A$Rs with gabazine selectively enhance nAChR-evoked axonal calcium signals but does not alter the strength or dynamics of acetylcholine release, suggesting that the enhancement is mediated primarily by $GABA_A$Rs on dopaminergic axons. Unexpectedly, we find that a widely used $GABA_A$R antagonist, picrotoxin, inhibits axonal nAChRs and should be used cautiously for striatal circuit analysis. Overall, we demonstrate that $GABA_A$Rs on dopaminergic axons regulate integration of nicotinic input to shape axonal excitability.

INTRODUCTION

Midbrain dopaminergic (DA) neurons contribute to reward processing and motor control by signaling through highly branched axons that densely innervate the striatum.[1,2] Several studies have demonstrated that various neurotransmitters are poised to locally modulate DA release via axonal receptors.[3–9] Among these receptors, ionotropic and metabotropic gamma-aminobutyric acid (GABA) receptors have been shown to influence DA release through multiple mechanisms, including via a striatal GABAergic tone.[10–16] Past work using direct axon recordings demonstrated that GABA type-A ($GABA_A$) receptors are present on the axons of DA neurons and can inhibit striatal DA release by decreasing axonal excitability and hindering action potential (AP) propagation.[14] However, how $GABA_A$ receptor signals interact with signaling from other axonal receptors remains an open question.

Nicotinic acetylcholine (ACh) receptors (nAChRs) on the axons of DA neurons are activated by release from striatal cholinergic interneurons (CINs), which provide another form of local control over striatal DA.[17–23] Synchronous activation of CINs causes robust DA release in *ex vivo* preparations.[19] Data from axonal recordings have shown that DA axons exhibit spontaneous, nAChR-mediated depolarizations as a result of transmission from nearby CINs.[22] While these individual depolarizing events occur at axonal membrane potentials below the AP threshold voltage ("subthreshold"), summation of these depolarizations in the axon can evoke axonal APs and drive DA release.[22,24,25] Thus, the ability to regulate the amplitude of nicotinic depolarizations would provide a means of controlling the excitability of DA axons and subsequent neurotransmitter release.

Previous work has shown that the $GABA_A$ positive allosteric modulator (PAM) diazepam reduces evoked DA release moderately in the presence of nAChR antagonists but inhibits DA release to a much larger extent when nicotinic transmission is left intact.[14] Although the mechanism is unclear, this observation raises the possibility that interactions between $GABA_A$ receptors and nAChRs may shape DA release. In particular, it is well known

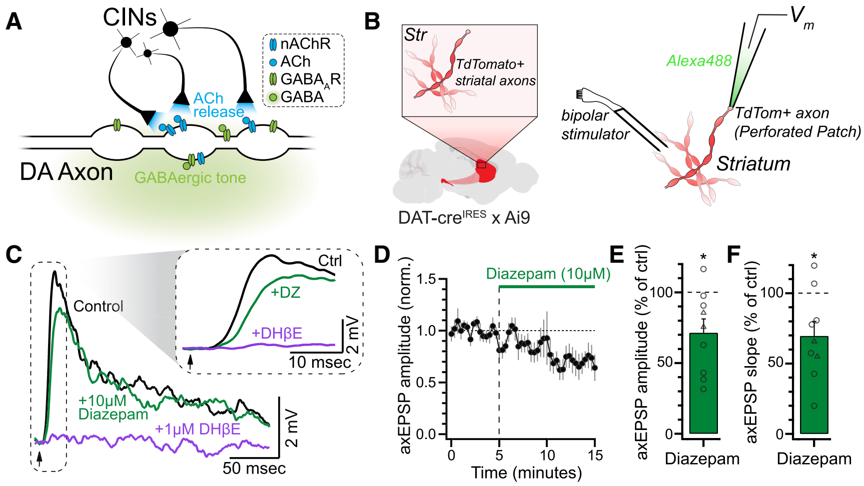

Figure 1. Diazepam suppresses nAChR-evoked axonal EPSPs recorded from dopaminergic axons

(A) Illustration of the simplified striatal environment that DA axons are subjected to, namely, a GABAergic tone that acts upon GABA$_A$ receptors and ACh release from CINs that act on nAChRs.

(B) Schematic of recording setup. Low-intensity electrical stimulation was applied in the vicinity of an axon recorded in a perforated-patch configuration.

(C) Average electrically evoked axEPSPs from a representative axon during control conditions, after diazepam (DZ; 10 μM) wash-in, and after DHβE (1 μM) wash-in.

(D) Normalized amplitude of evoked axEPSPs during diazepam wash-in (n = 9).

(E) Quantification of data in (D).

(F) The effect of diazepam on the rising slope of axEPSPs. Slope was measured between 30% and 70% of the peak amplitude.

In a subset of recordings (circles in E and F), nAChRs were partially blocked with a sub-saturating concentration of hexamethonium chloride (200 μM) to ensure that the amplitude of axEPSPs remained below the action potential threshold. Data are represented as mean ± SEM. *$p < 0.05$ (t test).

that GABA$_A$ receptors on the soma and dendrites can dampen neuronal excitability and limit the integration of excitatory inputs.[26–30] Whether axonal GABA$_A$ receptors play a similar role to shape nAChR-mediated input onto DA axons has not been examined.

Here, we tested the influence of axonal GABA$_A$ receptors on evoked nicotinic input within DA axons. Using direct axonal recordings, we found that the GABA$_A$ receptor PAM diazepam reduced the amplitude and slowed the rise time of nAChR-evoked axonal excitatory postsynaptic potentials (EPSPs). We then made use of genetically encoded calcium (Ca^{2+}) indicators (GECIs) with fast on kinetics to record electrically evoked, multi-component Ca^{2+} influx in DA axons. We found that the first derivative of these signals allows for unambiguous quantification of two temporally distinct components: one arising from direct DA axon stimulation and the other mediated by nAChR activation. Application of the GABA$_A$ receptor antagonist gabazine selectively enhanced the nAChR-mediated component of Ca^{2+} influx but did not alter the strength or dynamics of acetylcholine release. The kinetics of the nAChR-mediated component suggest that activation of GABA$_A$ receptors also decreases the precision of spike initiation in DA axons. Lastly, we show that the commonly used GABA$_A$ receptor antagonist picrotoxin—but not gabazine—inhibits nAChRs at concentrations that are typically used in *ex vivo* physiology experiments. Altogether, we demonstrate that axonal GABA$_A$ receptors impose a brake on the integration of nicotinic input in DA axons to shape presynaptic Ca^{2+} signals and axonal excitability.

RESULTS

Potentiation of GABA$_A$ receptor activation with diazepam suppresses nicotinic receptor-mediated EPSPs in DA axons

Striatal DA axons receive tonic GABA$_A$ receptor input and phasic nAChR input via ACh release from CINs (Figure 1A).[13,14,22] Previous experiments conducted in the presence of nAChR antagonists to isolate the effects of GABA$_A$ receptors showed that evoked DA release is reduced by bath application of the GABA$_A$ receptor PAM diazepam.[14] Interestingly, diazepam results in stronger inhibition of DA release when nicotinic transmission is left intact, which suggests an interaction between GABAergic and cholinergic signaling. The specific mechanisms that underlie this phenomenon are not understood.

We examined the effects of diazepam on nAChR-evoked electrical signaling in DA axons. To do so, we performed direct recordings from DA axons in the dorsomedial striatum (DMS) in horizontal slices (Figure 1B). Perforated-patch recordings from striatal DA axons were used to examine electrically evoked CIN transmission and limit rundown of nicotinic input.[22,31] Consistent with our past work,[22] we found that low-intensity electrical stimulation resulted in nAChR-mediated depolarizations. These axonal EPSPs (axEPSPs) were blocked by the nAChR antagonist dihydro-β-erythroidine hydrobromide (DHβE; 1 μM; Figure 1C). Following wash-in of diazepam (10 μM), we observed that the amplitude of evoked axEPSPs was reduced by ∼28% (71.4% ± 9.8% of control, p = 0.019, n = 9; Figures 1C–1E). In addition, the rising slope of the axEPSPs was significantly slower in diazepam when compared to that of control (69.6% ± 10.3% of control, p = 0.019, n = 9; Figures 1C and 1F). Together, these data suggest that the amplitude and kinetics of nicotinic input onto individual striatal DA axons are controlled by the activation of axonal GABA$_A$ receptors.

Electrically evoked Ca^{2+} signals from DA axons contain temporally and pharmacologically distinct components

To further investigate interactions between GABA$_A$ receptors and nAChRs, we next tested the excitability of DA axons at the population level by measuring presynaptic Ca^{2+} signals. Specifically, we expressed the high-affinity jGCaMP8s[32] (abbreviated as GCaMP) in DA neurons via viral injections in DAT-Cre mice (Figure 2A). To record action potential (AP)-evoked Ca^{2+} signals in DA axons, we used single pulses of local, near-maximal

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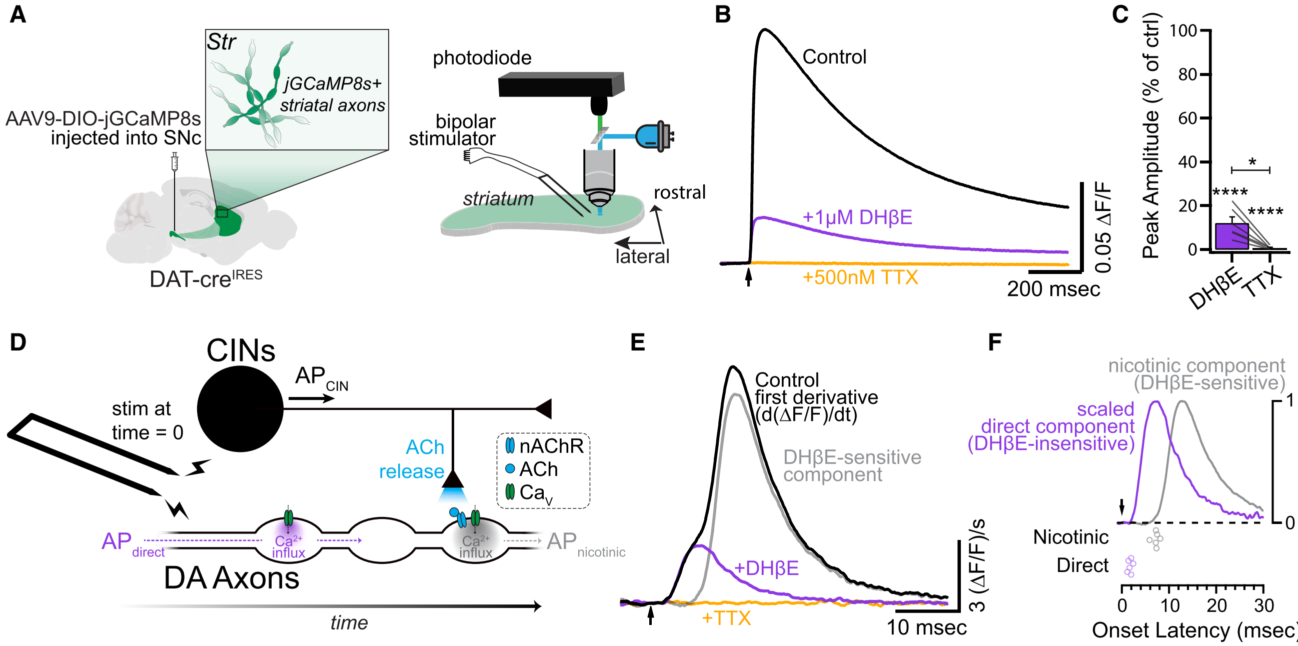

Figure 2. GCaMP as a readout of multi-component DA axon activation

(A) Schematic of GCaMP expression strategy and experimental setup. Horizontal slices containing the dorsomedial striatum were imaged with a photodiode while being stimulated with a bipolar electrode 200 μm away and out of the field of view of the photodiode.

(B) Fluorescent signals following a single electrical stimulus. Each trace represents an average of each drug condition in a representative experiment.

(C) Normalized amplitude in DHβE (1 μM) and TTX (500 nM) compared to control ($n = 6$).

(D) Schematic of CIN-DA axon circuitry and time course of APs following suprathreshold electrical stimulation.

(E) First derivative of traces in (B). The gray trace is the subtraction of the DHβE trace from the control trace.

(F) Top: normalized nicotinic and direct components from (E). Bottom: onset latencies for the two components ($n = 6$).

See also Figure S1. Data are represented as mean ± SEM. *$p < 0.05$ and ****$p \leq 0.0001$ (ANOVA).

electrical stimulation with bipolar electrodes placed in the striatum and measured fluorescence with a photodiode (Figure 2A). We found that antagonism of nAChRs with DHβE (1 μM) dramatically reduced the peak Ca^{2+} signal by ~88% compared to the original value (12.2% ± 2.8% of control, $p < 0.0001$, $n = 6$; Figures 2B and 2C). Subsequent addition of tetrodotoxin (TTX; 500 nM) completely abolished the remaining evoked Ca^{2+} signal (0.8% ± 0.3% of control, $p < 0.0001$, $n = 6$; Figures 2B and 2C). These data show that evoked Ca^{2+} signals represent axonal APs generated both by direct stimulation of DA axons and by axonal nAChR activation (Figure 2D).

To distinguish between nAChR-mediated and directly mediated Ca^{2+} influx during GCaMP transients in our photometric data, we calculated the first temporal derivative of the fluorescence signals, as previously done to estimate presynaptic Ca^{2+} currents in cerebellar parallel fibers.[33–35] The first derivative of the evoked DA axon Ca^{2+} signals revealed two components. The first "direct component" occurred immediately following electrical stimulation and was unchanged following blockade of nAChRs (Figures 2E and S1; see STAR Methods). This direct component was TTX sensitive and had an onset latency of 1.7 ± 0.2 ms ($n = 6$; Figures 2E and 2F). This component likely corresponds to the direct activation of DA fibers. The second, DHβE-sensitive "nicotinic component" occurred later in the signal and had an onset latency of 7.1 ± 0.3 ms ($n = 6$; Figure 2F). This latency measurement is in agreement with previous work that has shown that the latency of CIN transmission onto DA axons is around 7 ms.[22,36] The similarity of the waveforms of the two components is consistent with the notion that both reflect temporally distinct APs in DA axons (Figure 2F). Using these methods in subsequent experiments, therefore, enabled us to measure presynaptic Ca^{2+} influx resulting from both directly evoked APs and nAChR-evoked APs.

$GABA_A$ receptor antagonism enhances evoked nicotinic signals in DA axons

Using the derivative to distinguish between direct and nAChR-evoked DA axon activity, we next sought to examine how $GABA_A$ receptor activation interacts with evoked nicotinic transmission in the population of DA axons. One possibility is that axonal $GABA_A$ receptors could be regulating nAChR input to DA axons under control conditions. To test this, we bath applied the selective $GABA_A$ receptor antagonist SR-95531 (gabazine; 10 μM) following a baseline period. Gabazine application resulted in a significant increase of the electrically evoked GCaMP signal compared to control (115.6% ± 5.2% of control, $p = 0.02$, $n = 8$; Figures 3A–3C). Taking the first derivative revealed that this potentiation of the raw GCaMP signal was primarily driven by a significant increase in the magnitude of the nicotinic component (121.7% ± 7.8% of control, $p = 0.03$, $n = 8$; Figures 3D–3F). In

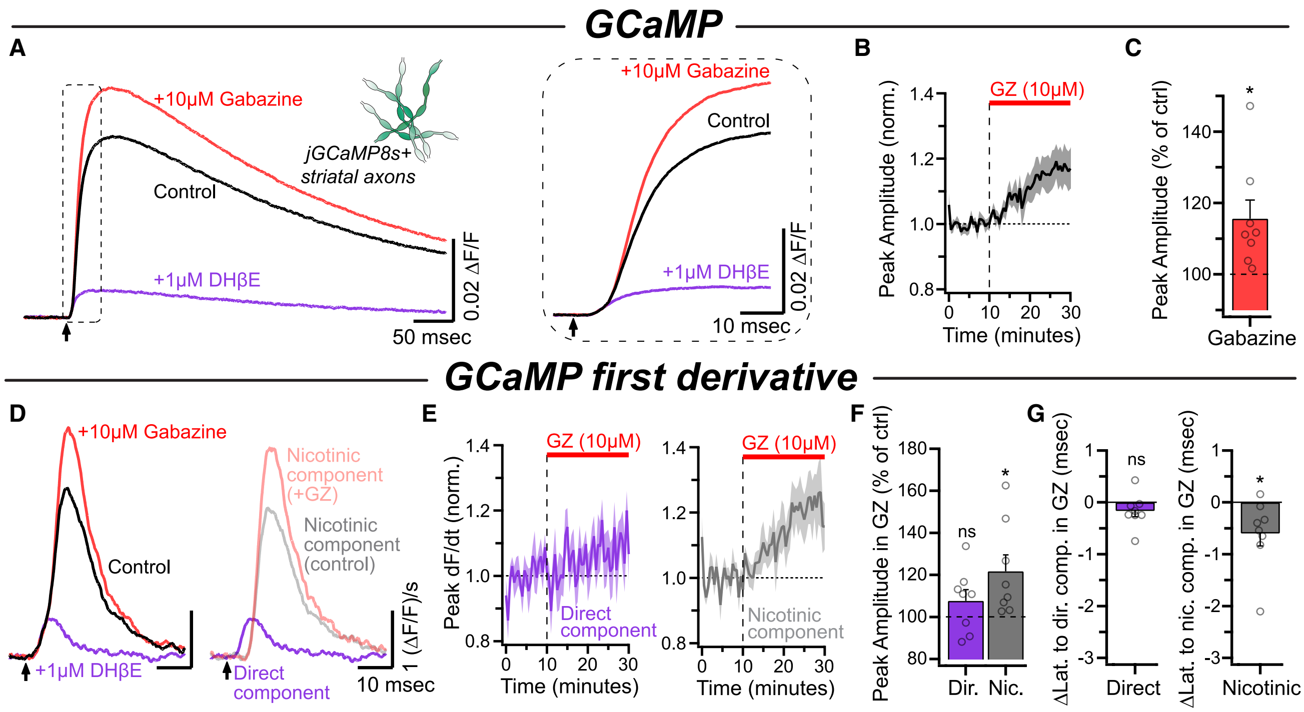

Figure 3. $GABA_AR$ antagonism potentiates the nicotinic component

(A) Average GCaMP signals for each condition in a representative experiment. The inset shows the same traces on an expanded timescale.

(B) Time course of the peak amplitude during gabazine (GZ; 10 μM) wash-in (n = 8).

(C) Quantification of the peak amplitude of the GCaMP signals in gabazine compared to control.

(D) Left: first derivative of traces in (A). Right: direct component and nicotinic components overlaid.

(E) Time course of normalized direct component and nicotinic component amplitude during wash-in of gabazine.

(F) Normalized amplitudes of both components in gabazine compared to control.

(G) Latency of the two components in gabazine compared to control.

See also Figure S2. Data are represented as mean ± SEM. $^*p < 0.05$ and non-significant (ns) $p > 0.05$ (t test).

contrast, the direct component exhibited a small but non-significant increase in magnitude (107.7% ± 5.3% of control, p = 0.19, n = 8; Figures 3E and 3F). This result is in agreement with our previous finding that blocking $GABA_A$ receptors has only a modest effect on local, electrically evoked DA release in the presence of nAChR antagonists.[14] Additionally, treatment with gabazine resulted in a shorter latency to peak for the nicotinic component, without affecting onset latency of the direct component (direct: −0.16 ± 0.11 ms, p = 0.19; nicotinic: −0.59 ± 0.24 ms, p = 0.043; n = 8; Figures 3G and S2).

The changes to the nicotinic component in gabazine could be explained by disinhibition of CINs, resulting in an increase in ACh release. To test this hypothesis, we examined the effect of gabazine directly on ACh release from CINs first using the fluorescent ACh sensor $GRAB_{ACh3.0}$[37] (abbreviated as $GRAB_{ACh}$) virally expressed on DA axons (Figure 4A). Under the same conditions as the experiments described above, we found that blockade of $GABA_A$ receptors with gabazine had no effect on the evoked $GRAB_{ACh}$ signal (97.5% ± 2.2% of control, p = 0.3, n = 6; Figures 4B and 4C). We next reasoned that if CIN $GABA_A$ receptors did have direct effects on ACh release that were not observable in the $GRAB_{ACh}$ experiments, then examining short-term plasticity of release may be a more sensitive readout. To test the effects of gabazine on CIN release dynamics, we measured axonal Ca^{2+} signals in response to a single electrical stimulus or a 5-pulse stimulus train (100 Hz; Figure 4D). Gabazine did not impact the 5p/1p ratio (Figure 4E), indicating that under these conditions, the effects observed above cannot be attributed to $GABA_A$ receptors on CINs. Together, these data suggest that $GABA_A$ receptors located on DA axons suppress nicotinic input in a manner that is independent of altering presynaptic release from CINs.

Picrotoxin inhibits nicotinic receptors on DA axons

Previous work has largely used two $GABA_A$ receptor antagonists, gabazine and picrotoxin, to probe the effects of $GABA_A$ receptor activation. To test the individual contributions of the two drugs, we applied picrotoxin to slices already bathed in gabazine. In contrast to gabazine alone, which enhanced the Ca^{2+} signals (Figure 3), we were surprised to see that subsequent application of picrotoxin (100 μM) resulted in a robust decrease in the evoked signal (62.5% ± 3.9% of control, p < 0.0001, n = 8; Figures 5A–5C). Comparison of the nicotinic and direct components of the Ca^{2+} signal derivative revealed that the effects of picrotoxin were isolated to the nicotinic component (direct: 106.3% ± 3.6% of control, p = 0.35; nicotinic: 45.8% ± 4.5% of control, p < 0.0001, n = 8; Figures 5D–5F). Additionally,

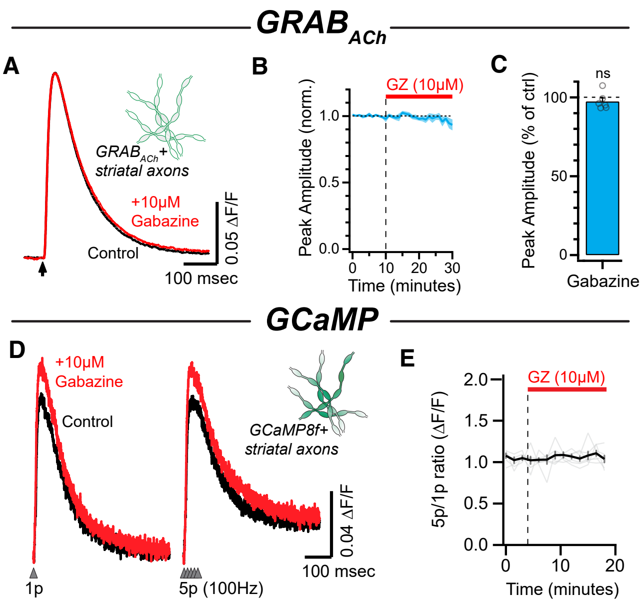

Figure 4. Changes to presynaptic ACh release from CINs cannot account for the effects of gabazine

(A) Average evoked $GRAB_{ACh}$ signals during control conditions and following wash-in of gabazine for a representative experiment.

(B) Time course of the normalized amplitude of $GRAB_{ACh}$ signals during gabazine wash-in ($n = 6$).

(C) Normalized amplitude of $GRAB_{ACh}$ signals in gabazine compared to control.

(D) Single-pulse and 5-pulse (100 Hz) electrically evoked Ca^{2+} signals in DA axons, before and after gabazine.

(E) Time course of the 5p/1p ratio during gabazine wash-in ($n = 6$). No change was observed.

Data are represented as mean ± SEM. ns $p > 0.05$ (t test).

this inhibition of evoked nicotinic input was dose dependent, with an IC_{50} of ~108 μM (95% confidence interval: 83.9–144.3 μM; Figure 5G). To examine if these effects resulted from a reduction in presynaptic ACh release, we imaged $GRAB_{ACh}$ in slices already bathed in gabazine (Figure S3A). The addition of picrotoxin to the bath solution did not influence the amplitude of the evoked $GRAB_{ACh}$ signals (99.7% ± 4.1% of control, $p = 0.94$, $n = 6$; Figures S3B and S3C), indicating that ACh release was not affected by picrotoxin. Together, these results indicate that picrotoxin acts on DA axons to suppress nicotinic Ca^{2+} influx, likely via direct nAChR inhibition.

To more directly test the effects of both picrotoxin and gabazine on axonal nAChRs, we made whole-cell axonal recordings from DA axons in the medial forebrain bundle (MFB; Figure 6A). The MFB DA axons were used in place of striatal DA axons to minimize the influence of endogenous GABAergic or cholinergic signaling,[14,22] allowing us to rule out circuit interactions as possible confounding variables. To activate nAChRs, we applied ACh (300 μM) using pressure ejections from a pipette positioned ~50 μm from the axonal recording site (Figure 6A), which produced time-locked depolarizations (Figures 6B and E). Application of gabazine (10 μM) did not significantly impact the amplitude of the ACh-evoked depolarization, whereas subsequent wash-on of DHβE abolished the response (ctrl vs. GZ, $p = 0.74$; GZ vs. DHβE, $p = 0.01$; mixed-effects model; $n = 6$; Figures 6B–6D and S4). By contrast, picrotoxin (100 μM) reliably reduced the amplitude of the ACh-evoked depolarization by ~50% (ctrl: 10.5 ± 0.6 mV; Ptx: 5.3 ± 0.5 mV, $p = 0.0018$; $n = 6$; Figures 6E–6G), consistent with a direct effect on nAChRs. Subsequent application of DHβE further reduced the depolarizations to amplitudes indistinguishable from baseline fluctuations in membrane potential (DHβE: 1.5 ± 0.2 mV, $p = 0.0008$, $n = 6$; Figures 6E–6G). Together, these results demonstrate that picrotoxin, but not gabazine, directly inhibits nAChRs on the axons of DA neurons.

$GABA_A$ and α6-containing nicotinic receptors produce opposing effects on timing of the nicotinic Ca^{2+} component

We next investigated whether pharmacological manipulations that modulate the amplitude and kinetics of axEPSPs can similarly influence the amplitude and latency of nicotinic-evoked Ca^{2+} signals. In the experiments above, we showed that diazepam potentiation of axonal $GABA_A$ receptors decreased the amplitude and slowed the kinetics of nAChR-evoked subthreshold depolarizations (Figure 1). Additionally, we showed that gabazine speeds the latency and increases the amplitude of the nicotinic component (Figure 3), suggesting again that $GABA_A$ receptors inhibit and slow nicotinic receptor-evoked input under control conditions. Therefore, our data suggest that the kinetics of the Ca^{2+} signals may reflect the underlying axEPSP waveforms that are only otherwise observable via axonal voltage recordings. This is likely due to changes in the ability of individual axEPSPs to integrate and trigger axonal APs.

To further test if the waveform of the nicotinic component of Ca^{2+} signals can reflect changes to axEPSPs, we applied conotoxin-P1A (P1A) to slices and recorded DA axon Ca^{2+} signals. P1A is a nAChR antagonist with high selectivity for α6 subunit-containing receptors,[38,39] which contribute to nAChR input to DA axons.[9,22,40,41] Our past work showed that P1A both slows axEPSP rise time and reduces its amplitude.[22] Together, these effects should reduce the ability of axEPSPs to summate and trigger APs. As expected, P1A (300 nM) reduced the amplitude of raw Ca^{2+} signals in DA axons to 30.2% ± 5.5% of control ($p < 0.001$, $n = 5$; Figures 7A–7C). Analysis of the derivative showed an ~85% decrease in the nicotinic component with no change to the direct component (Nic: 14.8% ± 3.3% of control, $p = 0.0001$; direct: 99.2% ± 1.7% of control, $p = 0.68$; $n = 5$; Figures 7D–7F). Subsequent application of picrotoxin further diminished the amplitude of the nicotinic component, from 14.8% in P1A to 2.7% of the control in picrotoxin (Figure S5). We also observed that P1A application resulted in a longer latency to peak of the nicotinic component (+2.8 ± 0.3 ms, $p < 0.001$, $n = 5$; Figures 7G and 7H) without altering the timing of the direct component (+0.1 ± 0.07 ms, $p > 0.99$, $n = 4$; Figures S6D and S6E). Thus, inhibition of α6 subunit-containing nAChRs, which slows axEPSP kinetics and decreases axEPSP amplitude,[22] also leads to smaller and more delayed nicotinic-evoked Ca^{2+} signals.

Interestingly, picrotoxin (applied after gabazine; data are from the same experiments as those shown in Figure 5) also delayed the time to peak of the nicotinic component, albeit to a lesser extent than P1A (1.5 ± 0.3 ms compared to control, $p < 0.001$, $n = 8$; Figures S6A–S6C). As with P1A, no change to the direct

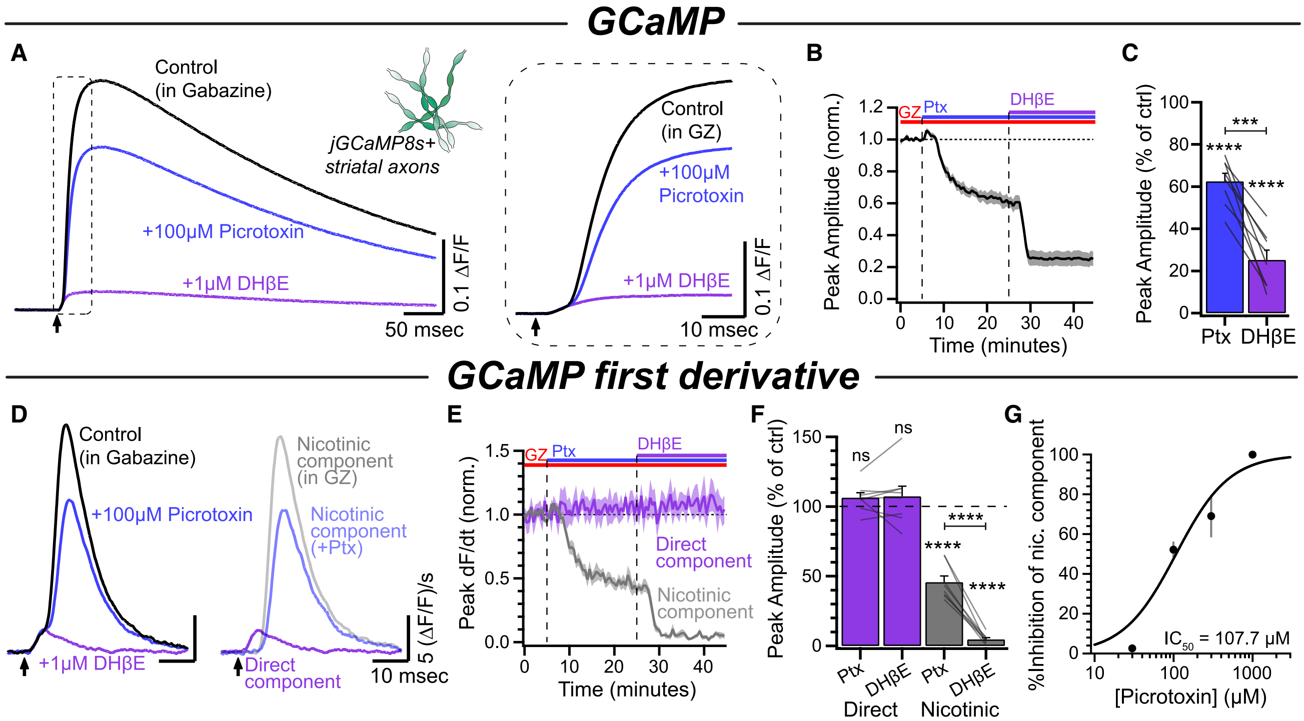

Figure 5. Picrotoxin inhibits nAChRs in a dose-dependent manner

(A) Average GCaMP signals for each condition in a representative experiment. The inset shows the same traces on an expanded timescale.

(B) Time course of the peak amplitude during picrotoxin (Ptx; 100 µM) and DHβE (1 µM) wash-in ($n = 8$).

(C) Quantification of the peak amplitude of the GCaMP signals in picrotoxin and DHβE compared to control (gabazine, 10 µM).

(D) Left: first derivative of traces in (A). Right: direct component and nicotinic components overlaid.

(E) Time course of normalized direct component and nicotinic component amplitude during wash-in of picrotoxin and DHβE.

(F) Normalized amplitudes of both components in picrotoxin and DHβE compared to control.

(G) Dose-response curve of the nicotinic component to increasing picrotoxin concentration (30 µM: $n = 3$; 100 µM: $n = 11$; 300 µM: $n = 5$; 1,000 µM: $n = 4$).

See also Figure S3. Data are represented as mean ± SEM. ***$p \leq 0.001$, ****$p \leq 0.0001$, and ns $p > 0.05$ (ANOVA).

component latency was detectable in picrotoxin (−0.09 ± 0.2 ms compared to control, $p > 0.99$, $n = 8$; Figures S6D and S6E). We examined the relationship between the nicotinic component amplitude and latency across the three drugs used. The changes in latency were negatively correlated with changes in amplitude across pharmacological conditions ($R^2 = 0.79$; Figure 7J). We interpret this as an indication that inhibition from $GABA_A$ receptor activation has a qualitatively similar effect to partial blocking of nAChRs on both axEPSPs and nicotinic component Ca^{2+} signals (i.e., smaller amplitude and slower kinetics). This is supported by the change to the axEPSP waveform in diazepam (Figure 1). Altogether, these data demonstrate that perturbations to either nAChR or $GABA_A$ receptor activation can alter both the magnitude and the timing of nicotinic-evoked Ca^{2+} influx in DA axons. Moreover, these data suggest that $GABA_A$ receptor activation impairs the ability of axEPSPs to summate and trigger APs in the axon, much like partial nAChR antagonism.

DISCUSSION

Here, we show that axonal $GABA_A$ receptors modulate the efficacy of nicotinic signaling onto DA axons, effectively gating cholinergic-evoked axon excitability. Using axonal voltage recordings, we show that the broad-spectrum benzodiazepine (BZD) diazepam dampens nAChR-mediated axEPSPs recorded from DA axons. This effect occurs through potentiation of axonal $GABA_A$ receptors that are activated by ambient GABA in the striatal slice. In photometry experiments, inhibition of $GABA_A$ receptors by gabazine enhanced nAChR-evoked Ca^{2+} signals in DA axons. Gabazine had no effect on evoked striatal ACh release, indicating that nicotinic-evoked Ca^{2+} signals are reduced primarily by axonal $GABA_A$ receptors on DA axons. Lastly, we provide evidence that picrotoxin—but not gabazine—exerts off-target effects and directly inhibits nAChRs on DA axons at concentrations often used in slice electrophysiology experiments. Together, these results add to our growing understanding of how axonal receptors control the physiology of DA axons and suggest that these axons continuously integrate local information arriving from the striatal circuitry.

First derivative of Ca^{2+} signals as a proxy for presynaptic Ca^{2+} influx and axon excitability

In our analysis of nAChR-evoked Ca^{2+} signals in DA axons, we used the first derivative of photometric Ca^{2+} signals to isolate

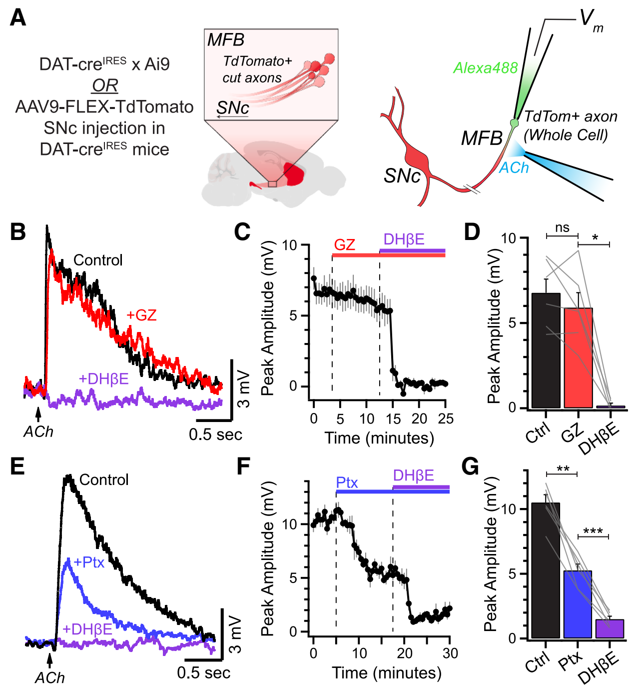

Figure 6. Picrotoxin, but not gabazine, decreases ACh-evoked depolarizations in the main axon of DA neurons

(A) Schematic of recording configuration. ACh (300 μM) was applied from a puff pipette positioned ~50 μm away from the recording site.

(B) Average ACh-evoked depolarizations in the main axon of a DA neuron before gabazine wash-in, in gabazine, and in DHβE.

(C) Time course of depolarization amplitude during wash-in of gabazine (10 μM) followed by DHβE.

(D) Amplitude of ACh-evoked depolarization in response to GZ and DHβE (n = 6).

(E) Same as (B) but with picrotoxin instead of gabazine.

(F) Time course of experiments with wash-in of picrotoxin and DHβE.

(G) Amplitude quantification for picrotoxin experiments (n = 6).

See also Figure S4. Data are represented as mean ± SEM. $*p < 0.05$, $**p \leq 0.01$, $***p \leq 0.001$, and ns $p > 0.05$ (D: mixed-effects model; G: ANOVA).

temporally distinct components of axonal Ca^{2+} influx. Previous work showed that the first derivative of fluorescent transients from low-affinity Ca^{2+}-sensitive fluorophores provides a quantitative measurement of the presynaptic Ca^{2+} current.[33–35] Similar derivative analyses have been applied to amperometric recordings of DA release.[36,41] However, an important advantage of imaging over amperometry is that Ca^{2+} signals are unaffected by stimulation artifacts. As a result, events that occur within a short latency relative to stimulation, such as direct DA axon activation, are more easily quantifiable when using optical approaches.

We show that jGCaMP8s has a sufficiently fast on rate to resolve millisecond-scale changes in both the kinetics and amplitude of Ca^{2+} influx in DA axons. Extending this analysis to other neuronal populations could provide valuable insights into axonal excitability and reveal how axonal receptors influence the physiology of presynaptic compartments across different neural circuits. More generally, this analysis of Ca^{2+} signal derivatives can be applied to imaging data recorded in a variety of experiments, including widely used *in vivo* photometry approaches. This approach can be particularly useful for distinguishing intrinsic AP firing from synaptically evoked Ca^{2+} signals obtained during photometry measurements made across cell compartments, including in axons and dendrites, as well as the soma.

Off-target effects of picrotoxin on DA axon nAChRs

Nicotinic receptors belong to the Cys-loop family of ligand-gated ion channels, along with $GABA_A$, glycine, and type-3 serotonin receptors.[42,43] While shared effects of antagonists such as bicuculline have been described,[44–46] the effects of picrotoxin on nAChRs are less clear and have been shown to differ across studies. Picrotoxin has been shown to inhibit α3β4- and α7-subunit-containing nAChRs in expression systems and guinea pig myenteric neurons,[47,48] while studies in rat embryonic skeletal muscle and mouse thalamic relay neurons show either weak or no effect on nAChR-mediated currents.[49,50] Using GCaMP photometry and direct axonal recordings, we show that nAChRs on DA axons are robustly inhibited by picrotoxin. Striatal DA axon terminals primarily express α4β2 and α6β2 nAChRs,[51–53] which indicates that picrotoxin likely inhibits β2-subunit-containing nAChRs. In addition, we found that picrotoxin leads to nAChR inhibition both in the absence and in the presence of P1A, which selectively blocks α6 subunit-containing nAChRs, suggesting multiple nAChR subunits may be targeted by picrotoxin. Thus, our findings suggest that picrotoxin is likely a nonselective inhibitor among the nicotinic receptors expressed on DA axons. Importantly, we find that gabazine (10 μM) has no measurable effect on nAChR activation in DA axons, in contrast to a previous report in locust neurons.[54] Thus, our data show that picrotoxin inhibits DA axon nAChRs, while gabazine is more specific and exhibits little to no off-target effects on these same receptors.

Although the off-target effects of picrotoxin in the striatum may complicate the interpretation of experiments, there are many conditions in which the use of picrotoxin can be straightforward. For example, picrotoxin can yield unambiguous results when used in the presence of blockers of nicotinic transmission.[13,14] Furthermore, optical stimulation of DA axons can significantly reduce the impact of picrotoxin's off-target nAChR inhibition when measuring evoked DA axon Ca^{2+} signals or evoked DA release. In this case, experiments can be designed to successfully mitigate the effects of picrotoxin that occur as a result of local striatal circuitry and altered tonic nAChR signaling.[16] Overall, our data suggest that the use of a more selective $GABA_A$ receptor antagonist, such as gabazine, is preferable for studies in the striatum or in other circuitry that involves both GABAergic and nicotinic signaling.

$GABA_A$ receptor-mediated control of DA axon excitability

Our results are in agreement with previous reports of a striatal GABAergic tone[10,55–57] and highlight the complex interplay between various neuronal populations in regulating striatal dopamine release. In experiments testing the effects of gabazine on Ca^{2+} signals, we show that $GABA_A$ receptor activation inhibits the nicotinic component of Ca^{2+} signals but has a minimal impact on the direct component. The latter part of this finding

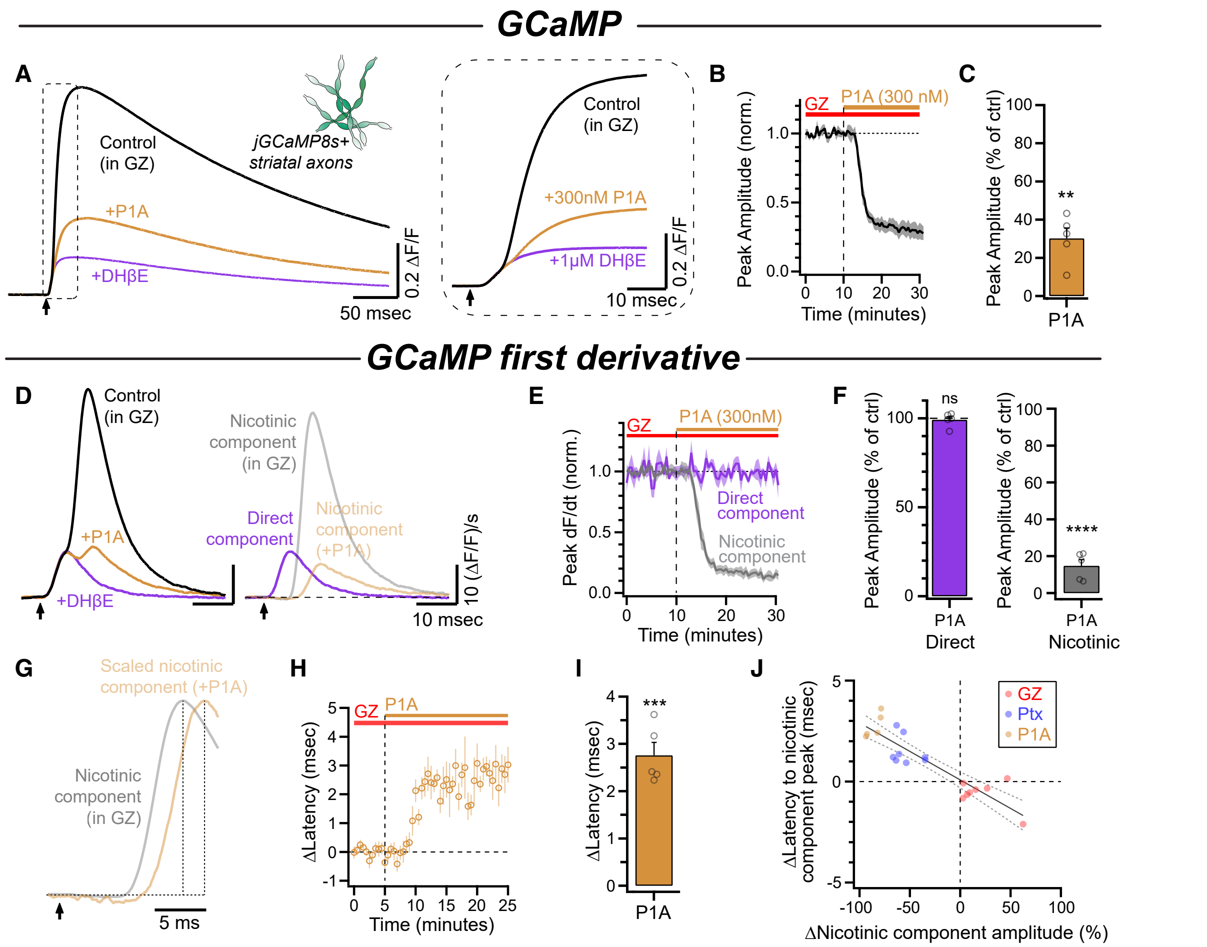

Figure 7. Conotoxin-P1A diminishes and slows the nicotinic component Ca^{2+} signal

(A) Average GCaMP signals for each condition in a representative experiment. The inset shows the same traces on an expanded timescale.

(B) Time course of the peak amplitude during conotoxin-P1A (300 nM) wash-in ($n = 5$).

(C) Quantification of the peak amplitude of the GCaMP signals in P1A compared to control (gabazine, 10 μM).

(D) Left: first derivative of traces in (A). Right: direct component and nicotinic components overlaid.

(E) Time course of normalized direct component and nicotinic component amplitude during wash-in of P1A.

(F) Normalized amplitudes of both components in P1A compared to control.

(G) Example normalized nicotinic component waveform before (in GZ) and after application of P1A. Note the increased latency in P1A.

(H) Time course of the time to peak of the nicotinic component following application of P1A ($n = 5$).

(I) Quantification of data shown in (H). No changes were observed in the onset latency of the direct component in the same experiment.

(J) Correlation of effect on nicotinic component amplitude and latency for gabazine, picrotoxin, and P1A (GZ: $n = 8$; Ptx: $n = 8$; P1A: $n = 5$; $y = -0.028x$; $R^2 = 0.79$). See also Figures S5 and S6. Data are represented as mean ± SEM. $^{**}p \leq 0.01$, $^{***}p \leq 0.001$, $^{****}p \leq 0.0001$, and ns $p > 0.05$ (t test).

is consistent with past work that found little effect of gabazine on evoked dopamine release when tested in the presence of nAChR antagonists.[14] Additionally, because CINs innervate local striatal GABAergic neurons and can robustly drive GABA release,[58] extracellular GABA is likely higher when the longer-latency nicotinic component occurs compared to the short-latency direct component and may therefore exert a stronger effect on the nicotinic component. The higher sensitivity of the nicotinic component to $GABA_A$ signaling likely also reflects axonal $GABA_A$ receptors on DA axons being well positioned to gate axonal nAChR input. Specifically, nAChR-generated APs in DA axons require integration of axEPSPs. Both functional and morphological data suggest that DA axons receive nicotinic input from release sites spread along the entire stretch of the axonal arbor.[22,23,59,60] Therefore, subthreshold axEPSPs are likely integrated across both time and space in order to summate and evoke axonal APs. These features may make nAChR-evoked activity in DA axons especially sensitive to inhibition from $GABA_A$ receptors.

$GABA_A$ receptor activation will influence integration through both changes in conductance ("shunting") and inactivation of voltage-gated sodium channels.[14] Differences in membrane conductance during APs and between spikes may also help explain why $GABA_A$ receptors have a stronger effect on nicotinic-evoked signals than on direct signals. During the time period between APs, where relatively few conductances are active and the axon is close to its resting potential, the input resistance of the axon is typically >1 GΩ.[14] During an axonal AP, however, axonal conductance can increase to 10–100 times its resting value.[61,62] Thus, $GABA_A$ receptor-mediated shunting will likely have a stronger effect on axEPSPs that occur near resting voltages, where the input resistance is high relative to that during APs. Simultaneously, the depolarization caused by activation of $GABA_A$ receptors will inactivate voltage-gated sodium channels on DA axons.[14] This would be expected to reduce boosting of axEPSP amplitude[22] and shorten the amplitude of axonal APs.

The specific sources of ambient GABA in our experiments are not well understood but likely include spontaneous release from spiny projection neurons, GABAergic interneurons, and even co-release from DA axons.[15,16,63–68] In addition, the electrical stimulation used in our experiments leads to evoked GABA release from these same cells and likely contributes to the effects observed. Interestingly, recent work showed that GABA co-released from DA axons can activate axonal $GABA_A$ receptors in an autoregulatory feedback loop.[16] It is also known that CINs form extensive connections with multiple cell types, including GABAergic interneurons,[58,69,70] and can directly evoke GABA release from DA axons.[71] Thus, the data presented here, combined with past results showing that CINs can drive GABA release, suggest that the CIN-DA axon striatal circuit is endowed with complex mechanisms for controlling striatal DA release, which includes negative feedback.

Our experiments using the acetylcholine sensor $GRAB_{ACh}$ showed that gabazine had no effect on ACh signals, suggesting that evoked transmission from CIN axons is not inhibited by striatal GABA under our conditions. Consistent with a lack of effect on presynaptic ACh release, gabazine did not impact the short-term plasticity of release from CINs, as measured by Ca^{2+} signals in DA axons. These results fit with the hypothesis that the inhibition of DA axons that we observed is mainly due to $GABA_A$ receptors located on DA axons rather than presynaptic $GABA_A$ receptors located on CIN axons. With that said, $GABA_A$ receptors on the cell bodies and dendrites of CINs can control ACh release by modulating firing,[58,69,72] which likely plays an additional role *in vivo*. In addition, our collective data from this and previous work[14] suggest that $GABA_A$ receptor-mediated inhibition of DA release occurs both due to a reduction of the intrinsic excitability of DA axons and due to diminished nicotinic depolarization of DA axons, suggesting an even more robust form of control than previously described.[13,14]

Benzodiazepine regulation of striatal dopamine

Benzodiazepines have long been known to modulate DA signaling, and a range of hypotheses have been suggested.[10,14,73–75] BZDs have been proposed to increase DA release via disinhibition of midbrain DA neurons.[73,76] Additionally, other work has shown that BZDs applied directly within the striatum reduce DA release, suggesting that local mechanisms are at play.[10,74,75] Our work here provides a mechanism for diazepam-induced reduction in striatal dopamine, which includes the potentiation of axonal $GABA_A$ receptors and the uncoupling of DA axons from nicotinic activation. This reduction of striatal dopamine could contribute to the lowered arousal and sedative properties that are commonly associated with BZDs.

Our previous findings,[14] in addition to those described in Figure 1, show that DA axons express functional BZD-sensitive $GABA_A$ receptors. Consistent with these data, RNA sequencing in DA neurons shows expression of several BZD-sensitive subunits, including γ2, α1, and α3 subunits.[77–80] A recent study also showed histological evidence for the α3 subunit on TH-positive processes in the striatum.[16] Together, these histological, transcriptional, and functional findings all corroborate each other. In future experiments, it will be important to determine the location of receptor subunits in different DA neuron subcompartments, such as the soma, dendrites, and axon terminals.

Our data also offer a possible explanation for the different effects of diazepam on DA release with and without nAChR antagonists that we reported previously.[14] With nicotinic transmission intact, the majority of electrically evoked DA release and evoked presynaptic Ca^{2+} in DA axons arises from nAChR activation and subsequent APs (Figure 2).[9,19,81] Our previous results explain the moderate effects of diazepam in the pharmacologically isolated condition and highlight the direct influence of $GABA_A$ receptors on DA axon excitability.[14] Our findings suggest that $GABA_A$ receptors additionally regulate DA axon excitability via control of nicotinic input, as illustrated by the decreased amplitude and slowed kinetics of axEPSPs in diazepam and the larger nicotinic component in gabazine. These data indicate that diazepam will also decrease nAChR-evoked DA release, leading to the larger effect of diazepam on DA release we observed with nicotinic signaling intact.[14]

DA axons as integrators of striatal information

Presynaptic receptors are poised to modulate DA release from striatal DA axons.[9,14,17–20,22,82] While it is well known that DA axons relay somatodendritic information via propagated APs, there is strong emerging evidence of an additional role of DA axons through local, modulatory control of DA release. Specifically, the recent description of spontaneous nicotinic axEPSPs and nAChR-evoked APs has provided a mechanistic explanation for how these receptors are able to shape axonal physiology in *ex vivo* slices.[22,23] These findings suggest that DA axons may also function in ways similar to dendrites, such as exhibiting synaptically evoked depolarizations that influence axonal excitability. Our results provide evidence for yet another function of DA axons, revealing their capacity to integrate inhibition and excitation in a manner reminiscent of the somatodendritic region. This points to the intriguing possibility that $GABA_A$ receptor activation could control how much influence CINs have over local DA release. Specifically, the local GABAergic tone could dictate the extent to which CIN synchrony is necessary to evoke APs in DA axons.

Importantly, changes to tonic GABAergic inhibition in the striatum have been reported in mouse models of neurodegenerative diseases, namely, Parkinson's disease (PD) and Huntington's disease.[66,68,83] For example, a recent study showed that in the absence of cholinergic influence, tonic GABAergic inhibition of DA axons (mainly via $GABA_B$ receptors) is substantially increased exclusively in certain striatal regions of an early-disease model of PD.[68] The interplay between these disease-related changes and well-documented changes to CINs in PD could be an interesting avenue of investigation, with potential therapeutic value.[84–87]

Here, we provide evidence that DA axons in the striatum integrate multiple inputs from their local environment. Specifically, the combination of $GABA_A$ and nicotinic input contributes to the output of DA axons via the regulation of local initiation of APs. Our data show that the effects of $GABA_A$ receptor activation on nicotinic input to DA axons are not due to circuit mechanisms. Instead, we propose that DA axons perform cell-intrinsic integration-like computations—a feature that has not previously been observed in axonal structures but is characteristic of postsynaptic dendrites. Additionally, we show that picrotoxin exerts robust off-target effects on nAChRs and should be used cautiously in the striatum and other complex circuitry. Finally, we describe a photometry analysis method that combines genetic specificity with high temporal resolution in order to study neuronal excitability. Overall, our data provide evidence for even more nuanced local control of DA axon physiology than previously understood, and we suggest an approach to study this phenomenon and similar ones across brain regions.

Limitations of the study

Electrical stimulation is a robust means of probing neuronal circuitry that does not rely on genetic control, nor does it interfere with optical methods for measuring activity. That said, it is also an inherently non-specific form of circuit activation. Our results are compatible with previous reports that have demonstrated the presence of a striatal GABAergic tone[10,13,14,55–57]; however, the experiments here cannot be used alone as evidence for such a phenomenon. That said, our use of predominantly single electrical stimuli (as opposed to trains or paired pulses of electrical stimuli) and our relatively long interstimulus intervals mean that the striatal GABAergic tone is likely a contributing factor in the modulation of DA axon excitability in these experiments.

Photometric measurements of cytoplasmic Ca^{2+} using GECIs can be a powerful approach to track neural activity in a cell-specific manner. However, this approach is limited in information regarding the cellular sources of Ca^{2+}.[88] For example, in our experiments, changes in magnitude to the Ca^{2+} signal following the application of drugs could result from a change in presynaptic Ca^{2+} influx within individual axons, a change in the number of axons being recruited to fire APs, or a combination of the two. In addition, subthreshold and AP-generated Ca^{2+} signals are not easily distinguishable. This limitation applies to all methods involving the measurement of bulk Ca^{2+} within a population of cells or axons. To address this issue, future work will require optical or electrical measurements from individual axons.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Zayd M. Khaliq (zayd.khaliq@nih.gov).

Materials availability

This study did not generate unique reagents.

Data and code availability

- The data reported in this paper have been deposited at 10.5281/zenodo.17308845 and are publicly available. The accession number is listed in the key resources table.
- This paper does not report original code.
- This paper does not report any other data or resources.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.G.B.-W. and Z.M.K.; methodology, S.G.B.-W., P.F.K., and Z.M.K.; investigation, S.G.B.-W., P.F.K., A.Y., A.M.L., F.H.C., and R.Z.; validation, S.G.B.-W., P.F.K., A.Y., A.M.L., and Z.M.K.; software, S.G.B.-W. and P.F.K.; formal analysis, S.G.B.-W. and Z.M.K.; data curation, S.G.B.-W., P.F.K., and Z.M.K.; visualization, S.G.B.-W. and Z.M.K.; resources, supervision, project administration, and funding acquisition, Z.M.K.; writing – original draft, S.G.B.-W.; writing – review & editing, S.G.B.-W. and Z.M.K., with feedback from all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
AAV9-pGP-*syn*-FLEX-jGCaMP8s-WPRE	Addgene	RRID:Addgene_162377
AAV9-pGP-*syn*-FLEX-jGCaMP8f-WPRE	Addgene	RRID:Addgene_162379
AAV9-hSyn-DIO-GRABACh3.0	Addgene	RRID:Addgene_121923
AAV9-CAG-Flex-TdTomato.WPRE.bGH	Penn Vector Core	RRID:Addgene_51503
Chemicals, peptides, and recombinant proteins		
Acetylcholine	Sigma	A6625
Conotoxin-P1a	Biomatic	N/A
dihydro-β-erythroidine hydrobromide (DHβE)	Tocris	2349
Diazepam	Sigma	D0899
Gramicidin	Sigma	G5002
(+)-Sodium L-ascorbate	Sigma	A4034
Sodium Pyruvate	Sigma	P5280
SR95531 hydrobromide (Gabazine)	Tocris	1262
Tetrodotoxin (TTX)	Tocris	1078
Thiourea	Sigma	T7875
Experimental models: Organisms/strains		
B6.SJL-Slc6a3(tm1.1(cre)Bkmn/J (DAT-IRES-cre)	The Jackson Laboratory	RRID:IMSR_JAX:006660
Ai9: Gt(ROSA)26Sor(tm9(CAG-tdTomato)Hze)	The Jackson Laboratory	RRID:IMSR_JAX:007909
Software and algorithms		
Igor Pro 9	Wavemetrics	RRID:SCR_000325
Prism	GraphPad	RRID:SCR_002798
Deposited data		
Data reported in main and supplemental figures	This paper	10.5281/zenodo.17308845

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mice

All animal handling and procedures were approved by the Animal Care and Use Committee (ACUC) for the National Institute of Neurological Disorders and Stroke (NINDS) at the National Institutes of Health. All experiments were performed on adult mice between the ages of p51 and p143 and mice of both sexes were used throughout the study. For patch clamp experiments, DAT-Cre (RRID:IMSR_JAX:006660) mice were crossed with Ai9 (RRID:IMSR_JAX:007909) mice and the resulting DAT-CreAi9 offspring were used. For imaging experiments, DAT-Cre mice underwent stereotaxic injections at postnatal day 39 or older and were used between 2 and 11 weeks later.

METHOD DETAILS

Viral injections

For stereotaxic injections, mice were maintained under isoflurane for the duration of the procedure. The viruses used for this study were jGCaMP8s (AAV9-pGP-*syn*-FLEX-jGCaMP8s-WPRE titer: > 1x 10^{13}; RRID:Addgene_162377), jGCaMP8f (AAV9-pGP-*syn*-FLEX-jGCaMP8f-WPRE titer: > 1x 10^{13}; RRID:Addgene_162379), $GRAB_{ACh3.0}$ (AAV9-hSyn-DIO-GRABACh3.0 titer: > 1x 10^{12}, plasmid: RRID:Addgene_121923, virus packaged by Vigene), and TdTomato (AAV9-CAG-Flex-TdTomato.WPRE.bGH; Penn Vector Core; RRID:Addgene_51503). Viral aliquots were injected (0.1–0.5 μL) bilaterally into the SNc (X: ±1.9 Y: −0.5 Z: −3.9) via a Hamilton syringe. At the end of the injection, the needle was raised at a rate of 0.1–0.2 mm per minute for 10 min before being removed. Mice were allowed to recover from anesthesia on a warmed pad and returned to the home cage.

Tissue slicing

Mice were anesthetized with isoflurane, decapitated, and brains were rapidly extracted. Horizontal sections were cut at 320–380 μm thickness on a vibratome while immersed in warmed (34°C), modified, slicing ACSF containing (in mM) 198 glycerol, 2.5 KCl, 1.2 NaH_2PO_4, 20 HEPES, 25 $NaHCO_3$, 10 glucose, 10 $MgCl_2$, 0.5 $CaCl_2$. Cut sections were promptly removed from the slicing chamber and incubated for 30–60 min in a heated (34°C) chamber with holding solution containing (in mM) 92 NaCl, 30 $NaHCO_3$, 1.2 NaH_2PO_4, 2.5 KCl, 35 glucose, 20 HEPES, 2 $MgCl_2$, 2 $CaCl_2$, 5 Na-ascorbate, 3 Na-pyruvate, and 2 thiourea. Slices were then stored at room temperature and used 30 min to 6 h after slicing. Following incubation, slices were moved to a heated (33°C–35°C) recording chamber that was continuously perfused with recording ACSF (in mM): 125 NaCl, 25 $NaHCO_3$, 1.25 NaH_2PO_4, 3.5 KCl, 10 glucose, 1 $MgCl_2$, 2 $CaCl_2$.

Electrophysiology and fluorescence imaging

For imaging experiments, a white light LED (Thorlabs; SOLIS-3C) was used in combination with a EGFP (Chroma; 49002) filter set to visualize DA axons infected with GCaMP or $GRAB_{ACh3.0}$. To visualize GCaMP or $GRAB_{ACh3.0}$ signals, a photodiode (New Focus) was mounted on the top port of the Olympus BX-51WI. Signals were acquired every 30 s (jGCaMP8s) or 60 s ($GRAB_{Ach}$) using a Digidata 1440A (Molecular Devices) sampled at 50kHz. For the short-term plasticity experiments (Figures 4D and 4E), jGCaMP8f signals were acquired with a 90 s inter-trial interval, and the burst stimulus was delivered 30 s following the single stimulus in each trial. For these experiments, the lower affinity GCaMP8 variant (jGCaMP8f) was used to avoid saturation during burst stimuli. All other GCaMP imaging experiments were done with jGCaMP8s. All electrical stimulation was delivered with tungsten bipolar electrodes (250 μm tip separation, MicroProbes) placed 200 μm from the imaging site in the DMS. Electrical stimulation was delivered using an Isoflex (A.M.P.I.) with amplitudes ranging from 0.3 to 0.6 V to produce a near-maximal (~90–95% of maximum) signal for photometry experiments, and a subthreshold voltage signal for perforated patch experiments.

Perforated-patch recordings from striatal TdTomato+ axons were made using borosilicate pipettes (6–9 MΩ) filled with internal solution containing (in mM) 135 KCl, 10 NaCl, 2 $MgCl_2$, 10 HEPES, 0.5 EGTA, 0.1 $CaCl_2$, adjusted to a pH value of 7.43 with KOH, 278 mOsm. Pipette tips were backfilled with ~1 μL of gramicidin-free internal. Pipettes were then filled with internal containing between 80 and 100 μg/mL gramicidin. Patch integrity was monitored by the addition of Alexa 488 to the gramicidin-containing internal. Whole-cell recordings from the TdTomato+ axons in the medial forebrain bundle were made using borosilicate pipettes (4–7 MΩ) filled with internal solution containing (in mM) 122 $KmeSO_3$, 9 NaCl, 1.8 $MgCl_2$, 4 Mg-ATP, 0.3 Na-GTP, 14 phosphocreatine, 9 HEPES, 0.45 EGTA, 0.09 $CaCl_2$, adjusted to a pH value of 7.35 with KOH. All recordings were made with a MultiClamp 700B (Molecular Devices).

Pressure ejection of acetylcholine was performed using borosilicate pipettes (2–4 MΩ). Acetylcholine (300 μM) was added to a modified external solution containing (in mM): 125 NaCl, 25 $NaHCO_3$, 1.25 NaH_2PO_4, 3.5 KCl, 10 HEPES, 0.01 Alexa 488, final osmolarity 280–290 mOsm. This puffing solution was then spin filtered, loaded into a glass pipette, and lowered to within 30–50 μm of the axon using a micro-manipulator. The puffing solution was applied onto the axon with a short pressure ejection (100–250 ms in duration) using a PV 820 Pneumatic PicoPump (WPI). In a subset of puffing experiments, TTX (500 nM) was included in the bath in order to eliminate APs propagating from the soma. In other cases, propagating APs were removed *post hoc* to aid in analysis of puff-evoked events.

QUANTIFICATION AND STATISTICAL ANALYSIS

Analysis was conducted in Igor Pro 9 (Wavemetrics; RRID:SCR_000325) and statistical tests were performed in Prism 9 (GraphPad; RRID:SCR_002798) and Igor Pro 9. Data in text are reported as the mean ± the standard error of the mean. T-tests were used for two-group comparisons, and ANOVA tests (unless otherwise noted) were used when comparing more than two groups followed by a Bonferroni post-hoc test to correct for multiple comparisons. In all figures, asterisks are used to denote corrected *p*-values as follows: *, $p < 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ****, $p \leq 0.0001$; ns, not significant before or after correction. In Figure S5F, $^{(*)}$, $p < 0.05$ before, but not after, correction.

axEPSP analysis

For amplitude measurements, evoked axEPSPs from every trial recorded were measured relative to baseline prior to the electrical stimulus. The data from each axon was normalized to its average amplitude during the control period. To perform measurements of axEPSP slope, average waveforms were generated for each axon in both control conditions and following application of diazepam. The rising slope of each average waveform was calculated from 30% to 70% of peak amplitude, and the slope of the diazepam waveform was normalized to that of the control waveform for each axon recorded.

Discrimination between direct and nicotinic components

We determined that the axonal Ca^{2+} signal was comprised on two components: a component resulting from the direct stimulation of dopaminergic axons ("direct component") and a delayed component that was dependent upon the activation of axonal nAChRs ("nicotinic component"). For discrimination between the direct and nicotinic components, analysis windows were defined based on average waveforms. Specifically, the direct component window was defined as a region ranging from 1.1–6.7 ms following the electrical stimulus. The window for the nicotinic component analysis was defined as a 20 ms window immediately following the direct component (6.7–36.7 ms after electrical stimulation; Figure S1).

Amplitude quantification of direct and nicotinic components

In order to quantify changes to the direct component of evoked GCaMP signals, the first derivative of the raw signal was lightly smoothed. The maximum value within the direct component analysis window was determined for every trace in each slice to create a time series. To quantify the nicotinic component, we determined the DHβE-sensitive component by subtracting the average waveform in DHβE from each control trace. To create the time series, the peak amplitude of each trace was measured from baseline and adjusted to account for baseline noise. Finally, both the quantifications of direct and adjusted nicotinic components for each slice were normalized to control conditions and averaged across slices. Quantification of drug effects was done by averaging the final 5 min of the control or wash-in periods.

Quantification of direct and nicotinic component kinetics

To find the onset latency of the direct component, the differentiated traces were used. Onset was defined for each trace as the point at which the trace crossed 10% of the maximum amplitude of the direct component. This point was identified by a backwards search beginning at the end of the direct component analysis window. The latency to peak of the nicotinic component was found by using the nicotinic component traces described above (Figure S2).